

Pakistan Navy Engineering College, Karachi
ational University of Sciences & Technology, Pakistan





EE 869 – Linear Systems Theory

Fall 2026

Lectures 7 – 8

Controllability

Motivating Example: A rocket in vertical motion may be modeled by

$$\begin{aligned} \dot{h} &= v \\ m\dot{v} &= -mg + f \end{aligned}$$

$h = \text{altitude},\ v = \text{velocity},\ m = \text{mass},\ f = \text{thrust force}$

$$x_1 = h,\ \ x_2 = v,\ \ u = f/m - g$$

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= u \end{aligned}$$

Can we find a (continuous) control $u(t)$ over the period $[t_0, t_f]$ to move the state of the system from a given initial state $x(t_0) = x_0$ to a desired final state $x(t_f) = x_f$?

Definition: The system $\dot{x} = Ax + Bu$, or the pair (A, B), is said to be controllable on $[t_0, t_f]$ if given any initial state x_0, there is a continuous control $u(t)$ that steers the state of the system from $x(t_0) = x_0$ to $x(t_f) = 0$. It is said to be reachable on $[t_0, t_f]$ if given any final state x_f, there is a continuous control $u(t)$ that steers the state of the system from $x(t_0) = 0$ to $x(t_f) = x_f$

Remark: For time-invariant systems, we can, without loss of generality, take $t_0 = 0$

Mathematical preliminaries: Let M be an $m \times n$ matrix

$$Mx = 0 \text{ for } x \neq 0 \quad \Rightarrow \quad \text{rank } M < n$$

$$y^T M = 0 \text{ for } y \neq 0 \quad \Rightarrow \quad \text{rank } M < m$$

If M is $n \times n$, then

$$M \text{ is nonsingular} \quad \Leftrightarrow \quad \text{rank } M = n$$

For a nonsingular matrix M

$$Mx = 0 \quad \Leftrightarrow \quad x = 0$$

$$x^T M = 0 \quad \Leftrightarrow \quad x = 0$$

A symmetric $n \times n$ matrix P is positive semidefinite if

$$x^T P x \geq 0, \quad \forall\, x$$

It is positive definite if

$$x^T P x > 0, \quad \forall\, x \neq 0$$

P is positive definite

$$\Leftrightarrow \quad \text{All eigenvalues of } P \text{ are positive}$$

$$\Leftrightarrow \quad \text{All leading principal minors of } P \text{ are positive}$$

A symmetric positive semidefinite matrix is positive definite if and only if it is nonsingular

Controllability

The Controllability Gramian of (A, B) is defined by

$$W_c(0, t_f) = \int_0^{t_f} e^{-At} B B^T e^{-A^T t} \, dt$$

$W_c(0, t_f)$ is symmetric. It is positive semidefinite because

$$x^T W_c(0, t_f) x = \int_0^{t_f} x^T e^{-At} B B^T e^{-A^T t} x \, dt \geq 0$$

for all vectors x.

Lemma: $W_c(0, t_f)$ is positive definite if and only if there is no vector $x_a \neq 0$ such that

$$x_a^T e^{-At} B \equiv 0, \quad \forall\, t \in [0, t_f]$$

Proof:

$$x_a^T W_c(0, t_f) x_a = \int_0^{t_f} x_a^T e^{-At} B B^T e^{-A^T t} x_a \, dt$$

$$x_a^T W_c(0, t_f) x_a = 0 \quad \Leftrightarrow \quad x_a^T e^{-At} B \equiv 0, \; \forall\, t \in [0, t_f]$$

Theorem: The pair (A, B) is controllable (reachable) on $[0, t_f]$ if and only if the controllability Gramian $W_c(0, t_f)$ is positive definite

Proof : Suppose $W_c(0, t_f)$ is positive definite and let x_0 be the initial state and x_f be the desired final state

$$x(t_f) = e^{At_f}x(0) + \int_0^{t_f} e^{A(t_f-\tau)}Bu(\tau)\, d\tau$$

Take

$$u(t) = B^T e^{-A^T t} W_c^{-1}(0, t_f)v$$

for some constant vector v.

$$x(t_f) = e^{At_f}x_0 + \int_0^{t_f} e^{A(t_f-\tau)}BB^T e^{-A^T \tau} W_c^{-1}(0, t_f)v\, d\tau$$

$$x(t_f) = e^{At_f}x_0 + e^{At_f}\int_0^{t_f} e^{-A\tau}BB^T e^{-A^T\tau}\,d\tau\, W_c^{-1}(0,t_f)v$$

$$x(t_f) = e^{At_f}x_0 + e^{At_f}W_c(0,t_f)W_c^{-1}(0,t_f)v$$

$$x(t_f) = e^{At_f}x_0 + e^{At_f}v = e^{At_f}(x_0 + v)$$

$$v = -x_0 + e^{-At_f}x_f \quad \Rightarrow \quad x(t_f) = x_f$$

Thus,

$$u(t) = B^T e^{-A^T t}W_c^{-1}(0,t_f)\left[-x_0 + e^{-At_f}x_f\right]$$

steers $x(0) = x_0$ to $x(t_f) = x_f$ for any x_0 and x_f

Rocket Example

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u = Ax + Bu$$

$$e^{At} = I + At = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$$

$$e^{-At}B = \begin{bmatrix} 1 & -t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -t \\ 1 \end{bmatrix}$$

$$W_c(0, t_f) = \int_0^{t_f} \begin{bmatrix} -t \\ 1 \end{bmatrix} \begin{bmatrix} -t & 1 \end{bmatrix} dt$$

$$W_c(0,t_f) = \int_0^{t_f} \begin{bmatrix} t^2 & -t \\ -t & 1 \end{bmatrix} dt = \begin{bmatrix} \frac{1}{3}t_f^3 & -\frac{1}{2}t_f^2 \\ -\frac{1}{2}t_f^2 & t_f \end{bmatrix}$$

Leading principal minors:

$$\frac{1}{3}t_f^3 > 0$$

$$\frac{1}{3}t_f^4 - \frac{1}{4}t_f^4 > 0$$

The system is controllable (reachable) on any interval $[0, t_f]$ with $t_f > 0$

Lemma: The controllability Gramian $W_c(0, t_f)$ is positive definite if and only if $\text{rank } \mathcal{C} = n$, where

$$\mathcal{C} = [B, AB, A^2B, \ldots, A^{n-1}B]$$

is the controllability matrix ($\mathcal{C}$ is $n \times nm$ when A is $n \times n$ and B is $n \times m$)

Proof:

$W_c(0, t_f)$ is singular if and only if

$$\text{there is } x_a \neq 0 \text{ such that } x_a^T e^{-At} B \equiv 0, \ \forall\, t \in [0, t_f]$$

$$\Leftrightarrow \quad x_a^T \sum_{k=0}^{\infty} \frac{(-t)^k}{k!} A^k B \equiv 0, \ \forall\, t \in [0, t_f]$$

$$\Leftrightarrow \quad \sum_{k=0}^{\infty} \frac{(-t)^k}{k!} x_a^T A^k B \equiv 0, \ \forall\, t \in [0, t_f]$$

$$\Leftrightarrow \quad x_a^T A^k B = 0, \ \text{for } k = 0, 1, 2, \ldots$$

By Cayley-Hamilton theorem, A satisfies its characteristic equation

$$\det(sI - A) = s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_n = 0$$

$$A^n + a_{n-1} A^{n-1} + \cdots + a_1 A + a_n I = 0$$

Hence, A^k, for $k \geq n$, are linear combinations of $I, \ A, \ \ldots, \ A^{n-1}$

$$x_a^T A^k B = 0,\ \forall\, k \geq 0 \Leftrightarrow x_a^T A^k B = 0,\ \text{for } k = 0, 1, 2, \ldots, n$$

$W_c(0, t_f)$ is singular if and only if

$$x_a^T [B, AB,\ A^2B,\ \ldots,\ A^{n-1}B] = 0$$

$$\Leftrightarrow \quad \text{rank}\,[B, AB,\ A^2B,\ \ldots,\ A^{n-1}B] < n$$

Theorem: The pair (A, B) is controllable (reachable) if and only if $\text{rank}\ \mathcal{C} = n$

Remark: Controllability of (A, B) is independent of the interval $[0, t_f]$

Example: Investigate controllability of

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\mathcal{C} = [B,\ AB] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\text{rank } \mathcal{C} = 2 \quad \Rightarrow \quad (A, B) \text{ is controllable}$$

Example: Investigate controllability of

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\mathcal{C} = [B,\ AB,\ A^2B] = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -3 \\ 1 & -3 & 7 \end{bmatrix}$$

$$\text{rank}\, \mathcal{C} = 3 \quad \Rightarrow \quad (A, B) \text{ is controllable}$$

Matlab:

$$C = ctrb(A, B); rank(C)$$

Example: Investigate controllability of

$$A = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}, \quad B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\mathcal{C} = [B,\ AB,\ A^2B] = \begin{bmatrix} b_1 & \lambda_1 b_1 & \lambda_1^2 b_1 \\ b_2 & \lambda_2 b_2 & \lambda_2^2 b_2 \\ b_3 & \lambda_3 b_3 & \lambda_3^2 b_3 \end{bmatrix}$$

$$\text{If } b_i = 0 \text{ for some } i,\ \text{rank}\,\mathcal{C} < 3$$

$$\text{If } \lambda_i = \lambda_j \text{ for some } i \neq j,\ \text{rank}\,\mathcal{C} < 3$$

$$\mathcal{C} = \begin{bmatrix} b_1 & \lambda_1 b_1 & \lambda_1^2 b_1 \\ b_2 & \lambda_2 b_2 & \lambda_2^2 b_2 \\ b_3 & \lambda_3 b_3 & \lambda_3^2 b_3 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} b_1 & \lambda_1 b_1 & \lambda_1^2 b_1 \\ 0 & (\lambda_2 - \lambda_1) b_2 & (\lambda_2^2 - \lambda_1^2) b_2 \\ 0 & (\lambda_3 - \lambda_1) b_3 & (\lambda_3^2 - \lambda_1^2) b_3 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} b_1 & \lambda_1 b_1 & \lambda_1^2 b_1 \\ 0 & (\lambda_2 - \lambda_1) b_2 & (\lambda_2^2 - \lambda_1^2) b_2 \\ 0 & 0 & * \end{bmatrix}$$

$$* = (\lambda_3^2 - \lambda_1^2) b_3 - (\lambda_2^2 - \lambda_1^2) \frac{(\lambda_3 - \lambda_1)}{(\lambda_2 - \lambda_1)} b_3$$

$$
\begin{aligned}
* &= (\lambda_3^2 - \lambda_1^2) b_3 - (\lambda_2^2 - \lambda_1^2) \frac{(\lambda_3 - \lambda_1)}{(\lambda_2 - \lambda_1)} b_3 \\
&= (\lambda_3 - \lambda_1) \left[(\lambda_3 + \lambda_1) - (\lambda_2 + \lambda_1) \right] b_3 \\
&= (\lambda_3 - \lambda_1)(\lambda_3 - \lambda_2) b_3
\end{aligned}
$$

$$
\mathcal{C} \rightarrow \begin{bmatrix} b_1 & \lambda_1 b_1 & \lambda_1^2 b_1 \\ 0 & (\lambda_2 - \lambda_1) b_2 & (\lambda_2^2 - \lambda_1^2) b_2 \\ 0 & 0 & (\lambda_3 - \lambda_1)(\lambda_3 - \lambda_2) b_3 \end{bmatrix}
$$

The pair (A, B) is controllable if and only if $b_i \neq 0$ for all i and $\lambda_i \neq \lambda_j$ for all $i \neq j$

Observability

Definition: The system

$$\dot{x} = Ax, \quad y = Cx$$

or the pair (A, C), is said to be observable on $[t_0, t_f]$ if any initial state $x(t_0) = x_0$ can be uniquely determined from $y(t)$ on $[t_0, t_f]$. It is said to be (re)constructible on $[t_0, t_f]$ if any final state $x(t_f) = x_f$ can be uniquely determined from $y(t)$ on $[t_0, t_f]$

Without loss of generality, take $t_0 = 0$

The observability Gramian of (A, C) is defined by

$$W_o(0, t_f) = \int_0^{t_f} e^{A^T t} C^T C e^{At} \, dt$$

Lemma: $W_o(0, t_f)$ is positive definite if and only if there is no vector $x_a \neq 0$ such that

$$Ce^{At}x_a \equiv 0, \quad \forall\, t \in [0, t_f]$$

Proof:

$$x_a^T W_o(0, t_f) x_a = \int_0^{t_f} x_a^T e^{A^T t} C^T C e^{At} x_a \, dt$$

$$x_a^T W_o(0, t_f) x_a = 0 \quad \Leftrightarrow \quad Ce^{At}x_a \equiv 0, \; \forall\, t \in [0, t_f]$$

Theorem: The pair (A, C) is observable (constructible) on $[0, t_f]$ if and only if the observability Gramian $W_o(0, t_f)$ is positive definite

Proof: Suppose $W_o(0, t_f)$ is positive definite and let x_0 be the initial state

$$y(t) = Cx(t) = Ce^{At}x_0$$

Multiply from the left by $e^{A^T t}C^T$

$$e^{A^T t}C^T y(t) = e^{A^T t}C^T Ce^{At}x_0$$

Integrate from 0 to t_f

$$\int_0^{t_f} e^{A^T t}C^T y(t)\, dt = \int_0^{t_f} e^{A^T t}C^T Ce^{At}\, dt\, x_0$$

$$\int_0^{t_f} e^{A^T t} C^T y(t)\, dt = W_o(0, t_f) x_0$$

This equation has a unique solution

$$x_0 = W_o^{-1}(0, t_f) \int_0^{t_f} e^{A^T t} C^T y(t)\, dt$$

Since

$$x(t_f) = e^{A t_f} x_0$$

and $e^{A t_f}$ is nonsingular, $x(t_f)$ is determined uniquely by

$$x(t_f) = e^{A t_f} W_o^{-1}(0, t_f) \int_0^{t_f} e^{A^T t} C^T y(t)\, dt$$

Lemma: The observability Gramian $W_o(0, t_f)$ is positive definite if and only if $\mathrm{rank}\ \mathcal{O} = n$, where

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

is the observability matrix ($\mathcal{O}$ is $np \times n$ when A is $n \times n$ and C is $p \times n$)

Proof:

$W_o(0, t_f)$ is singular if and only if

$$\text{there is } x_a \neq 0 \text{ such that } Ce^{At}x_a \equiv 0, \ \forall\, t \in [0, t_f]$$

$$\Leftrightarrow \quad C \sum_{k=0}^{\infty} \frac{t^k}{k!} A^k x_a \equiv 0, \ \forall\, t \in [0, t_f]$$

$$\Leftrightarrow \quad CA^k x_a = 0, \ \ \forall\, k \geq 0$$

$$\Leftrightarrow \quad CA^k x_a = 0, \ \ \forall\, k = 0, 1, \ldots, n-1$$

$$\Leftrightarrow \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} x_a = 0$$

$$\Leftrightarrow \text{rank} \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} < n$$

Theorem: The pair (A, C) is observable (reconstructible) if and only if $\text{rank } \mathcal{O} = n$

Duality

$$(A, B) \text{ controllable} \Leftrightarrow \text{rank}\,[B, AB, \ldots, A^{n-1}B] = n$$

$$\Leftrightarrow \text{rank} \begin{bmatrix} B^T \\ B^T A^T \\ \vdots \\ B^T (A^T)^{n-1} \end{bmatrix} = n$$

$$\Leftrightarrow \left(A^T, B^T\right) \text{ observable}$$

Similarly, observability of (A, C) is equivalent to controllability of $\left(A^T, C^T\right)$

Controllability and observability are invariant to state transformations

$$\{A, B, C, D\} \rightarrow \boxed{x = Pz} \rightarrow \{P^{-1}AP, P^{-1}B, CP, D\}$$

$$\begin{aligned}
\text{rank} \left[P^{-1}B,\ P^{-1}APP^{-1}B,\ \ldots,\ \left(P^{-1}AP\right)^{n-1} P^{-1}B \right] \\
&= \text{rank}\ \left[P^{-1}B,\ P^{-1}AB,\ \ldots,\ P^{-1}A^{n-1}B\right] \\
&= \text{rank}\ \left\{P^{-1}[B,\ AB,\ \ldots,\ A^{n-1}B]\right\} \\
&= \text{rank}\ [B,\ AB,\ \ldots,\ A^{n-1}B]
\end{aligned}$$

Matlab:

$$\mathcal{O} = obsv(A, C);\ \ rank(\mathcal{O})$$

Mathematical Preliminaries: Consider the equation

$$y = Mx$$

where M is an $m \times n$ matrix. The equation has a solution if and only if y can be expressed as a linear combination of the columns of M

$$\text{rank } M = \text{rank } [M,\ y]$$

The range space of M, denoted by $\mathcal{R}(M)$, is the set of all linear combinations of the columns of M. It is a subspace of R^m of dimension equal to $\text{rank } M$. The equation $y = Mx$ has a solution if and only if $y \in \mathcal{R}(M)$. The equation $y = Mx$ has a solution for every $y \in R^m$ if and only if

$$\text{rank}(M) = m \quad \Leftrightarrow \quad \mathcal{R}(M) = R^m$$

Given that $y = Mx$ has a solution. Is the solution unique?

Suppose there are two solutions x_a and x_b such that

$$y = Mx_a, \qquad y = Mx_b$$

$$M(x_a - x_b) = 0$$

The null space of M, denoted by $\mathcal{N}(M)$, is the set of all vectors $x \in R^n$ such that $Mx = 0$. It is a subspace of R^n of dimension $(n - \text{rank } M)$

The solution of $y = Mx$ is unique if and only if there is no vector $x \neq 0$ such that $Mx = 0$

Equivalently, $\text{rank } M = n$

Decomposition of Noncontrollable & Nonobservable Systems

Decomposition of Noncontrollable Systems

Theorem: Consider the system

$$\dot{x} = Ax + Bu$$

where A is $n \times n$ and B is $n \times m$

$$\text{Let} \qquad \mathcal{C} = [B,\ AB,\ \ldots\ ,\ A^{n-1}B]$$

If $\text{rank}\ \mathcal{C} = q < n$, then there is a nonsingular $n \times n$ matrix P such that

$$\hat{A} = P^{-1}AP = \begin{bmatrix} \hat{A}_{11} & \hat{A}_{12} \\ 0 & \hat{A}_{22} \end{bmatrix}, \quad \hat{B} = P^{-1}B = \begin{bmatrix} \hat{B}_1 \\ 0 \end{bmatrix}$$

where $\hat{A}_{11}$ is $q \times q$, $\hat{B}_1$ is $q \times m$, and the pair $(\hat{A}_{11}, \hat{B}_1)$ is controllable

Proof: Let

$$P = \begin{bmatrix} X & Y \end{bmatrix}$$

where X is an $n \times q$ matrix whose columns are linearly independent columns of $\mathcal{C}$ (or any q columns that form a basis of $\mathcal{R}(\mathcal{C})$) and Y is an $n \times (n-q)$ matrix whose columns are chosen such that P is nonsingular; i.e., all the columns of P are linearly independent

$$\mathcal{R}(X) = \mathcal{R}(\mathcal{C})$$

$$p \in \mathcal{R}(\mathcal{C}) \;\Rightarrow\; Ap \in \mathcal{R}(\mathcal{C}) \;\Rightarrow\; \mathcal{R}(AX) \subset \mathcal{R}(\mathcal{C})$$

Partition P^{-1} as $P^{-1} = \begin{bmatrix} M \\ N \end{bmatrix}$, where M is $q \times n$ and N is $(n-q) \times n$

$$P^{-1}P = I_n$$

$$\begin{bmatrix} M \\ N \end{bmatrix} \begin{bmatrix} X & Y \end{bmatrix} = \begin{bmatrix} MX & MY \\ NX & NY \end{bmatrix} = \begin{bmatrix} I_q & 0 \\ 0 & I_{n-q} \end{bmatrix}$$

$$NX = 0 \Rightarrow \mathcal{R}(\mathcal{C}) \subset \mathcal{N}(N)$$

$$\Rightarrow \quad NB = 0 \text{ and } NAX = 0$$

$$\hat{A} = P^{-1}AP = \begin{bmatrix} M \\ N \end{bmatrix} A \begin{bmatrix} X & Y \end{bmatrix} = \begin{bmatrix} MAX & MAY \\ NAX & NAY \end{bmatrix}$$

$$\hat{B} = P^{-1}B = \begin{bmatrix} M \\ N \end{bmatrix} B = \begin{bmatrix} MB \\ NB \end{bmatrix}$$

$$\hat{A} = \begin{bmatrix} \hat{A}_{11} & \hat{A}_{12} \\ 0 & \hat{A}_{22} \end{bmatrix}, \quad \hat{B} = \begin{bmatrix} \hat{B}_1 \\ 0 \end{bmatrix}$$

$$\hat{\mathcal{C}} = [\hat{B},\ \hat{A}\hat{B},\ \ldots,\ \hat{A}^{n-1}\hat{B}] = P^{-1}\mathcal{C}$$

$$\operatorname{rank} \hat{\mathcal{C}} = \operatorname{rank} \mathcal{C} = q$$

$$\hat{\mathcal{C}} = \left[\begin{array}{cccc|c} \hat{B}_1 & \hat{A}_{11}\hat{B}_1 & \cdots & \hat{A}_{11}^{q-1}\hat{B}_1 & \cdots & \hat{A}_{11}^{n-1}\hat{B}_1 \\ 0 & 0 & \cdots & 0 & \cdots & 0 \end{array}\right]$$

By Cayley-Hamilton theorem

$$\operatorname{rank} \hat{\mathcal{C}} = \operatorname{rank} \begin{bmatrix} \hat{B}_1 & \hat{A}_{11}\hat{B}_1 & \cdots & \hat{A}_{11}^{q-1}\hat{B}_1 \end{bmatrix} = q$$

$$\Rightarrow \quad (\hat{A}_{11}, \hat{B}_1) \text{ is controllable}$$

The change of variables

$$x = Pz = P \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

transforms the system into the form

$$\begin{aligned} \dot{z}_1 &= \hat{A}_{11} z_1 + \hat{A}_{12} z_2 + \hat{B}_1 u \\ \dot{z}_2 &= \hat{A}_{22} z_2 \end{aligned}$$

Clearly the state z_2 is not controllable

$$z_2(0) = 0 \;\Rightarrow\; z_2(t) \equiv 0 \;\Rightarrow\; \dot{z}_1 = \hat{A}_{11} z_1 + \hat{B}_1 u$$

The state z_1 is controllable because the pair $(\hat{A}_{11}, \hat{B}_1)$ is controllable

In the z-coordinates we can only steer states of the form

$$z = \begin{bmatrix} z_1 \\ 0 \end{bmatrix}$$

In the x-coordinates

$$x = Pz = \begin{bmatrix} X & Y \end{bmatrix} \begin{bmatrix} z_1 \\ 0 \end{bmatrix} = Xz_1$$

We can only steer states that belong to $\mathcal{R}(\mathcal{C})$. Hence, $\mathcal{R}(\mathcal{C})$ is called the controllable subspace

$$\hat{A} = P^{-1}AP = \begin{bmatrix} \hat{A}_{11} & \hat{A}_{12} \\ 0 & \hat{A}_{22} \end{bmatrix}$$

The eigenvalues of A and $\hat{A}$ are the same. Because $\hat{A}$ is block triangular, the eigenvalues of A are given by the eigenvalues of $\hat{A}_{11}$ and $\hat{A}_{22}$

The eigenvalues of $\hat{A}_{11}$ are called the controllable eigenvalues and those of $\hat{A}_{22}$ are the called the uncontrollable eigenvalues

Example: Decomposition of Noncontrollable System

$$A = \begin{bmatrix} 1 & 2 & -1 & -1 \\ -1 & -2 & 0 & -1 \\ -1 & -1 & 0 & 2 \\ 1 & 1 & -1 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\mathcal{C} = \begin{bmatrix} 1 & 3 & -5 & 5 \\ 1 & -4 & 5 & -4 \\ -1 & 0 & 1 & -2 \\ 1 & 0 & -1 & 2 \end{bmatrix}$$

$$\text{rank } \mathcal{C} = 3 \quad \Rightarrow \quad (A, B) \text{ is not controllable}$$

$$\operatorname{rank} \begin{bmatrix} 1 & 3 & -5 \\ 1 & -4 & 5 \\ -1 & 0 & 1 \\ 1 & 0 & -1 \end{bmatrix} = 3$$

The first three columns of $\mathcal{C}$ are linearly independent

$$P = \begin{bmatrix} 1 & 3 & -5 & * \\ 1 & -4 & 5 & * \\ -1 & 0 & 1 & * \\ 1 & 0 & -1 & * \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 3 & -5 & 0 \\ 1 & -4 & 5 & 0 \\ -1 & 0 & 1 & 0 \\ 1 & 0 & -1 & 1 \end{bmatrix}, \quad \operatorname{rank} P = 4$$

$$\hat{A} = P^{-1}AP = \begin{bmatrix} 0 & 0 & -1 & 1.5 \\ 1 & 0 & -3 & 5 \\ 0 & 1 & -3 & 3.5 \\ 0 & 0 & 0 & -1 \end{bmatrix}, \quad \hat{B} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\hat{A} = P^{-1}AP = \left[\begin{array}{ccc|c} 0 & 0 & -1 & 1.5 \\ 1 & 0 & -3 & 5 \\ 0 & 1 & -3 & 3.5 \\ \hline 0 & 0 & 0 & -1 \end{array}\right], \quad \hat{B} = \left[\begin{array}{c} 1 \\ 0 \\ 0 \\ \hline 0 \end{array}\right]$$

$$\hat{A}_{11} = \begin{bmatrix} 0 & 0 & -1 \\ 1 & 0 & -3 \\ 0 & 1 & -3 \end{bmatrix}, \quad \hat{B}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Matlab Calculation

```
>> A = [1 2 -1 -1;-1 -2 0 -1;-1 -1 0 2;1 1 -1 -3];
>> B = [1;1;-1;1];
>> SC = ctrb(A,B); rank(SC)
ans =
     3
>> rank(SC(:,1:2))
ans =
     2
>> rank(SC(:,1:3))
ans =
     3
>> P = [SC(:,1:3),[0;0;0;1]]; rank(P)
ans =
     4
>> Ah = inv(P)*A*P; Bh = inv(P)*B;
>> Ah_11 = Ah(1:3,1:3); Bh_1 = Bh(1:3);
>> rank(ctrb(Ah_11,Bh_1))
ans =
     3
```

Decomposition of Nonobservable Systems

Theorem: Consider the system

$$\dot{x} = Ax, \quad y = Cx$$

where A is $n \times n$ and C is $p \times n$

$$\text{Let} \qquad \mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

If $\text{rank}\, \mathcal{O} = q < n$, then there is a nonsingular $n \times n$ matrix P such that

$$\hat{A} = P^{-1}AP = \begin{bmatrix} \hat{A}_{11} & 0 \\ \hat{A}_{21} & \hat{A}_{22} \end{bmatrix}, \quad \hat{C} = CP = \begin{bmatrix} \hat{C}_1 & 0 \end{bmatrix}$$

where $\hat{A}_{11}$ is $q \times q$, $\hat{C}_1$ is $p \times q$, and the pair $(\hat{A}_{11}, \hat{C}_1)$ is observable

Proof: Let

$$P^{-1} = \begin{bmatrix} M \\ N \end{bmatrix}$$

where M is a $q \times n$ matrix whose rows are linearly independent rows of $\mathcal{O}$ and N is an $(n-q) \times n$ matrix whose rows are chosen such that P^{-1} is nonsingular; i.e., all the rows of P^{-1} are linearly independent

$$\mathcal{N}(M) = \mathcal{N}(\mathcal{O})$$

$$p \in \mathcal{N}(\mathcal{O}) \;\Rightarrow\; Ap \in \mathcal{N}(\mathcal{O}) \;\Rightarrow\; MAp = 0,\ \forall\, p \in \mathcal{N}(\mathcal{O})$$

Partition $P = \left(P^{-1}\right)^{-1}$ as $P = \begin{bmatrix} X & Y \end{bmatrix}$, where X is $n \times q$ and Y is $n \times (n-q)$

$$P^{-1}P = I_n$$

$$\begin{bmatrix} M \\ N \end{bmatrix} \begin{bmatrix} X & Y \end{bmatrix} = \begin{bmatrix} MX & MY \\ NX & NY \end{bmatrix} = \begin{bmatrix} I_q & 0 \\ 0 & I_{n-q} \end{bmatrix}$$

$$MY = 0 \;\Rightarrow \mathcal{R}(Y) = \mathcal{N}(\mathcal{O})$$

The columns of Y form a basis of $\mathcal{N}(\mathcal{O})$

$$\Rightarrow \quad CY = 0 \ \text{ and } \ MAY = 0$$

$$\hat{A} = P^{-1}AP = \begin{bmatrix} M \\ N \end{bmatrix} A \begin{bmatrix} X & Y \end{bmatrix} = \begin{bmatrix} MAX & MAY \\ NAX & NAY \end{bmatrix}$$

$$\hat{C} = CP = C \begin{bmatrix} X & Y \end{bmatrix} = [CX, \ CY]$$

$$\hat{A} = \begin{bmatrix} \hat{A}_{11} & 0 \\ \hat{A}_{21} & \hat{A}_{22} \end{bmatrix}, \quad \hat{C} = \begin{bmatrix} \hat{C}_1 & 0 \end{bmatrix}$$

$$\hat{\mathcal{O}} = \begin{bmatrix} \hat{C} \\ \hat{C}\hat{A} \\ \vdots \\ \hat{C}\hat{A}^{n-1} \end{bmatrix} = \mathcal{O}P, \ \ \text{rank}\, \hat{\mathcal{O}} = \text{rank}\, \mathcal{O} = q$$

$$\hat{\mathcal{O}} = \left[\begin{array}{cc} \hat{C}_1 & 0 \\ \hat{C}_1\hat{A}_{11} & 0 \\ \vdots & \vdots \\ \hat{C}_1\hat{A}_{11}^{q-1} & 0 \\ \hline \vdots & \vdots \\ \hat{C}_1\hat{A}_{11}^{n-1} & 0 \end{array}\right], \quad \text{rank}\,\hat{\mathcal{O}} = \text{rank}\left[\begin{array}{c} \hat{C}_1 \\ \hat{C}_1\hat{A}_{11} \\ \vdots \\ \hat{C}_1\hat{A}_{11}^{q-1} \end{array}\right] = q$$

$$\Rightarrow \quad (\hat{A}_{11}, \hat{C}_1) \text{ is observable}$$

Remark: We can also choose $P = \left[\begin{array}{cc} X & Y \end{array}\right]$, where the $(n-q)$ columns of Y form a basis of $\mathcal{N}(\mathcal{O})$ and the q columns of X are chosen such that P is nonsingular

The change of variables

$$x = Pz = P \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

transforms the system into the form

$$\begin{aligned} \dot{z}_1 &= \hat{A}_{11} z_1 \\ \dot{z}_2 &= \hat{A}_{21} z_1 + \hat{A}_{22} z_2 \\ y &= \hat{C}_1 z_1 \end{aligned}$$

Clearly the state z_2 is not observable, while z_1 is observable because the pair $(\hat{A}_{11}, \hat{C}_1)$ is observable

$$z_1(0) = 0 \quad \Rightarrow \quad z_1(t) \equiv 0$$

In the z-coordinates, the state

$$z = \begin{bmatrix} 0 \\ z_2 \end{bmatrix}$$

is unobservable
In the x-coordinates

$$x = Pz = \begin{bmatrix} X & Y \end{bmatrix} \begin{bmatrix} 0 \\ z_2 \end{bmatrix} = Y z_2 \in \mathcal{N}(\mathcal{O})$$

Any state in $\mathcal{N}(\mathcal{O})$ is unobservable. Hence, $\mathcal{N}(\mathcal{O})$ is called the unobservable subspace

$$\hat{A} = P^{-1}AP = \begin{bmatrix} \hat{A}_{11} & 0 \\ \hat{A}_{21} & \hat{A}_{22} \end{bmatrix}$$

The eigenvalues of $\hat{A}_{11}$ are called the observable eigenvalues and those of $\hat{A}_{22}$ are the called the unobservable eigenvalues

Kalman Decomposition Theorem: Consider the system

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx \end{aligned}$$

where A is $n \times n$, B is $n \times m$, and C is $p \times n$. Let

$$\begin{aligned} q_c &= \text{rank}\, \mathcal{C} \quad (\dim(\mathcal{R}(\mathcal{C})) = q_c) \\ q_o &= \text{rank}\, \mathcal{O} \quad (\dim(\mathcal{N}(\mathcal{O})) = n - q_o) \\ q_{c\bar{o}} &= \dim(\mathcal{R}(\mathcal{C}) \cap \mathcal{N}(\mathcal{O})) \quad (q_{c\bar{o}} \leq \min\{q_c, n - q_o\}) \end{aligned}$$

Then there is a nonsingular $n \times n$ matrix P such that $\hat{A} = P^{-1}AP$, $\hat{B} = P^{-1}B$, and $\hat{C} = CP$ are given by

$$\hat{A} = \begin{bmatrix} \hat{A}_{11} & 0 & \hat{A}_{13} & 0 \\ \hat{A}_{21} & \hat{A}_{22} & \hat{A}_{23} & \hat{A}_{24} \\ 0 & 0 & \hat{A}_{33} & 0 \\ 0 & 0 & \hat{A}_{43} & \hat{A}_{44} \end{bmatrix}, \quad \hat{B} = \begin{bmatrix} \hat{B}_1 \\ \hat{B}_2 \\ 0 \\ 0 \end{bmatrix}$$

$$\hat{C} = \begin{bmatrix} \hat{C}_1 & 0 & \hat{C}_3 & 0 \end{bmatrix}$$

$\hat{A}_{ii}$ is $q_i \times q_i$, for $i = 1, 2, 3, 4$, $\quad n = q_1 + q_2 + q_3 + q_4$

$$\left(\begin{bmatrix} \hat{A}_{11} & 0 \\ \hat{A}_{21} & \hat{A}_{22} \end{bmatrix}, \begin{bmatrix} \hat{B}_1 \\ \hat{B}_2 \end{bmatrix} \right) \text{ is controllable}$$

$$\left(\begin{bmatrix} \hat{A}_{11} & \hat{A}_{13} \\ 0 & \hat{A}_{33} \end{bmatrix}, \begin{bmatrix} \hat{C}_1 & \hat{C}_3 \end{bmatrix} \right) \text{ is observable}$$

$$
\begin{aligned}
q_2 &= q_{c\bar{o}} \\
q_1 &= q_c - q_{c\bar{o}} \\
q_4 &= n - q_o - q_{c\bar{o}}
\end{aligned}
$$

The eigenvalues of $\hat{A}_{11}$ are controllable and observable

The eigenvalues of $\hat{A}_{22}$ are controllable and unobservable

The eigenvalues of $\hat{A}_{33}$ are uncontrollable and observable

The eigenvalues of $\hat{A}_{44}$ are uncontrollable and unobservable

If $q_i = 0$ for $i = 1,\ 2,\ 3,\ \text{or}\ 4$, the corresponding part of the decomposition disappears

Proof: Read the textbook

The matrix P is taken as

$$P = \begin{bmatrix} P_1 & P_2 & P_3 & P_4 \end{bmatrix}$$

where P_i has q_i columns, for $i = 1, 2, 3, 4$

The columns of $\begin{bmatrix} P_1 & P_2 \end{bmatrix}$ form a basis of $\mathcal{R}(\mathcal{C})$

The columns of P_2 form a basis of $\mathcal{R}(\mathcal{C}) \cap \mathcal{N}(\mathcal{O})$

The columns of $\begin{bmatrix} P_2 & P_4 \end{bmatrix}$ form a basis of $\mathcal{N}(\mathcal{O})$

The columns of P_3 are chosen such that P is nonsingular

Thank You